

E-commerce Platform Search Function Analysis

Comparison of Linear Search and Binary Search

Linear Search:

Working

Linear Search checks each product one by one until it finds the required product ID.

Time Complexity

Best Case: $O(1)$ If value was matched in the first iteration.

Average Case: $O(n)$

Worst Case: $O(n)$

Binary Search

Working

Binary Search repeatedly divides the sorted array into halves.

Time Complexity

Best Case: $O(1)$ If the middle was the target.

Average Case: $O(\log n)$

Worst Case: $O(\log n)$

Which Algorithm is More Suitable?

Linear Search

Advantages:

- Easy to implement.
- No sorting required.

Disadvantages:

- Slow for large datasets.

Binary Search

Advantages:

- Very fast.
- Efficient for large product catalogs.
- Scales well as the number of products increases.

Disadvantages:

- Requires the products to be sorted.

Conclusion

For an e-commerce platform, **Binary Search is more suitable** because:

1. Product databases may contain thousands or millions of products.
2. Binary Search has **$O(\log n)$** complexity, making searches much faster.
3. Products can be sorted once and searched many times efficiently.